FINAL PROJECT SCENARIO 1

Understanding Diabetes Progression

A complete data analysis project in R

Project overview

In this project, you will investigate a real health dataset containing measurements from 442 patients. The dataset includes a measure of diabetes progression one year after baseline together with 10 patient characteristics, including age, sex, body mass index, blood pressure and several serum measurements.

You will carry out a complete data analysis workflow in R. The project begins with understanding and exploring the data, moves through relationships and regression modelling, and ends with a careful discussion of uncertainty, causality and limitations.

The supplied starter R file contains ready-to-run code and extensive comments for the required analysis. Your task is to run and understand the code, make appropriate choices where asked, and interpret the results in your own words.

Main research question: Which patient characteristics are associated with diabetes progression, what do the regression models tell us, and what conclusions can reasonably be drawn from the data?

Dataset

Dataset	Diabetes Progression
Observations	442 patients
Outcome	Continuous measure of disease progression one year after baseline
Predictors	Age, sex, BMI, blood pressure and six serum measurements
Analysis software	R
Source	NCSU-hosted copy of the diabetes dataset

Dataset file: diabetes.tab.txt (supplied with the project). Original public source: https://www4.stat.ncsu.edu/~boos/var.select/diabetes.tab.txt

The six project questions

Your analytical report should be organised around the six questions below. You do not need to reproduce every piece of R output. Select the results, tables and figures that help you answer each question clearly.

Question 1. Understand and prepare the data

- What does the dataset contain, and what does one row represent?
- Identify the outcome and the main predictor variables. Note that a variable may be coded with numbers but still represent categories. In this dataset, sex should be treated as categorical rather than as a continuous numerical measurement.
- Import the data into R and inspect its dimensions, names and data types.
- Check for missing values, duplicated rows and obviously problematic values.
- Briefly explain why privacy, responsible use and cautious interpretation matter when working with health-related data.

Answer in the report: What data are available, and are there any important data-quality or data-use issues?

Question 2. Explore the data

- Calculate suitable descriptive statistics for the outcome and important predictors. Use measures such as mean, median, standard deviation, range and interquartile range where useful.
- Create a visualisation of diabetes progression and a visualisation of BMI.
- Create at least one additional visualisation that helps you understand the dataset.
- Comment briefly on distributions, spread, unusual observations and other patterns that matter for later analysis.

Answer in the report: What are the most important characteristics of the dataset before modelling begins?

Question 3. Investigate relationships

- Use a correlation matrix or heatmap to investigate relationships among the numerical measurement variables. Do not include sex in the Pearson correlation matrix, because it is treated as a categorical variable in this project.
- Create at least two scatterplots showing potentially interesting relationships with diabetes progression.
- Identify the variables that appear most strongly associated with progression.
- Check whether some predictors are strongly related to each other and could create multicollinearity problems later.

Answer in the report: Which variables appear most promising for further analysis, and why?

Question 4. Build and interpret regression models

- Fit one simple linear regression model using a meaningful predictor. BMI is a natural starting point, but you may choose another predictor if you justify your choice.
- Interpret the fitted equation, slope and R-squared in the context of the data.
- Fit one multiple regression model using a small set of predictors informed by your exploratory analysis.
- Interpret the main coefficients using the idea of holding the other predictors constant.

- Compare the simple and multiple models. Do not judge the models from R-squared alone.

Answer in the report: What do the simple and multiple regression models tell us about diabetes progression?

Question 5. Evaluate the analysis

- Inspect a residuals-versus-fitted plot and comment on any clear warning signs.
- Check for possible multicollinearity using the correlation information from Question 3 and the VIF code supplied in the starter file.
- For at least two important coefficients, use the estimate together with a confidence interval, t-value or p-value to discuss statistical uncertainty.
- Distinguish statistical significance from practical importance.

Answer in the report: How much confidence should we have in the model results, and what are the main statistical weaknesses or uncertainties?

Question 6. Draw appropriate conclusions

- Explain whether the dataset is experimental or observational and why this matters for causal interpretation.
- Discuss possible confounding variables, missing information or measurement issues.
- Consider whether the results would necessarily generalise to another population.
- Summarise the conclusions that are supported by the analysis and identify claims that would go beyond the evidence.

Answer in the report: What can and cannot reasonably be concluded from this analysis?

Required deliverables

1. Analysis file

Submit an R Markdown, Quarto document, R notebook, or well-organised R script containing the code required to reproduce your analysis. The code should have clear sections, meaningful names and brief comments where useful. The supplied starter R file may be used and adapted.

2. Analytical report

Submit a 4 to 6 page PDF report, excluding references or appendices. Organise the report around the six project questions. Use selected tables and figures to support your answers, but do not paste complete R output into the report.

3. Decision-maker presentation

Prepare a separate presentation of 4 to 6 slides for a health-service decision-maker who understands the general idea of data analysis but is not a statistician. This is a communication task, not a second analytical report.

- Present the main findings and their practical meaning.

- Explain the main uncertainty or limitation in accessible language.
- State clearly what should not be concluded from the analysis.
- Use a small number of well-chosen figures or results. Avoid raw statistical output and unnecessary technical notation.

Suggested report structure

1. Introduction and data
2. Exploratory analysis
3. Relationships between variables
4. Regression modelling
5. Model evaluation and uncertainty
6. Causality, limitations and conclusions

Important guidance

- The purpose of the project is not to reproduce a predetermined numerical answer. Your results should come from your own analysis in R.
- Do not treat a statistically significant association as evidence of causation.
- Do not judge a model only from one metric. Consider fit, uncertainty, assumptions, interpretation and practical usefulness together.
- Use the starter R file as a scaffold. It contains working code, but you are responsible for understanding the steps and interpreting the results.
- A strong report explains what the results mean rather than copying raw software output.
- The presentation is a separate communication task. It should summarise the analysis for a non-specialist audience rather than repeat the report slide by slide.